

NEWKer



Making the most ideal and practical
product to service CNC world

NEW15/18iM Modbus type(EtherCAT)

Milling machine controller with machining center

Installation Guide

CHENGDU NEWKer CNC-TECHNOLOGY CO.,LTD

controller

NEWKER PRODUCTS LIST

990 series(2-4 axis)

990T series

990M series

990TDCa 990TDCb 16iT 990MDCa 990MDCb 16iM

T: lathe machine M: milling machine Ca incremental(pulse+directon) Cb absolute(pulse+direction+RS485) i(modbus)

1000 series

1000T series

1000M series

1000TDCa 1000TDCb 18iT 1000MDCa 1000MDCb 18iM

1000TDC/1000MDC 2-5 axis

18iT/18iM 2-8 axis

1500 series

1500T series

1500M series

1500TDCa 1500TDCb 15iT 1500MDCa 1500MDCb 15iM

axis support same as 1000 series

Dual channel controller

robot arm controller

22iT/M(total max 16 axis)

NEWKer i series(4-12 axis)

Optional (machining center ATC for milling machine, special program)

1000MiCa/b 1500MiCa/b 18iMi

drive

DS series incremental drive

DS301 DS302 DS501 DS503 DS753

NEW series absolute drive

NEW301 NEW302 NEW501 NEW503 NEW753

Modbus drive

NK301 NK302 NK501 NK503 NK753

Motor

incremental motor 2500PPR incremental encoder

60 series 80 series 90 serie 110 series 130 series 150 series 180 series

absolute motor 17bit absolute encoder

60 series 80 series 90 serie 110 series 130 series 150 series 180 series

Configuration

incremental controller + incremental drive + incremantal motor

absolute controller + absolute drive + absolute motor

+ incremental drive + incremantal motor

Modbus controller + modbus drive + absolute motor

+ absolute drive + absolute motor

+ incremental drive + incremantal motor

Robot arm

MODEL	JOINTS	PAYLOAD(kgs)	RANGE(mm)	<u>positioner</u>
NKRM40814	4	8	1400	<u>welding machine</u>
NKRM42017	4	20	1700	<u>spindle (servo drive+motor)</u>
NKRM418031	4	180	3100	<u>transformer</u>
NKRT60407	6	4	700	
NKRT1506B	6	6	1500	
NKRT1506A	6	6	1500	
NKRT1806A	6	6	1800	
NKRT1510A	6	10	1500	
NKRT2020A	6	20	2000	
NKRT1830A	6	30	1800	
NKRT1550A	6	50	1500	
NKRT2560A	6	60	2500	
NKRT2380A	6	80	2300	
NKRT627210	6	270	2100	

1.NEWKer Model

NEWKER 数控
鑫科瑞数控 CNC Milling Controller € €

Model: NEW1000MDCa-4

Input Power: ~220V 50Hz Power: 100W

Serial Number: 20101000.13.149

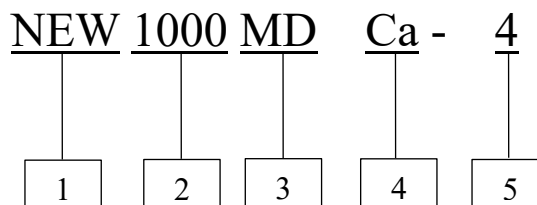
Ex-factory Date: OCT,2020 **Note!**

1. The system must be reliably grounded!
 2. Must be powered by an isolation transformer.

CHENGDU NEWKER CNC-TECHNOLOGY CO.,LTD

MADE IN CHINA

➤ Model Explanation



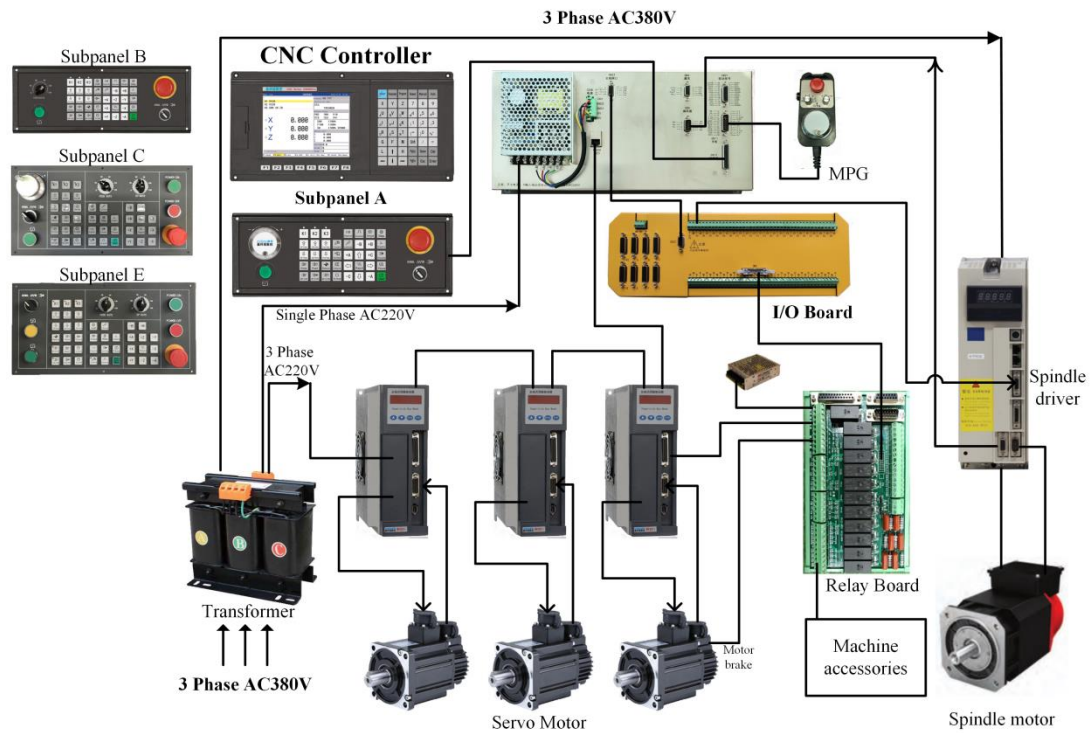
No.	Item	Description	Model meaning
1	Plant code	NEW	NEWKer product
2	Series code	990: integrated panel model; 1000: Standard panel model; 1500: Vertical panel model; 16i: Bus type integrated panel model; 18i: Bus type standard panel model; 15i:Bus type vertical panel model;	1000 series standard panel model
3	Application	TD: Lathe MD: Milling machine Mi: Milling machine with ATC function(1000/1500 series) T: Bus type for Lathe M: Bus type for milling machine Mi: bus type for milling machine with ATC function(18/15 series)	Milling controller
4	Function	Ca: incremental function(990/1000/1500 series) Cb: absolute function(990/1000/1500 series)	Incremental function
5	Axis number	2: 2 axis(XZ) 3: 3 axis(XYZ/XZC) 4: 4 axis(XYZA/XZCA) 5: 5 axis(XYZAB/XZCAB) 6: 6 axis(XYZABC/XZYABC) 7: 7 axis(XYZABCXs/XZYABCXs) 8: 8 axis(XYZABCXsYs/XZYABCXsYs)	4 axis(XYZA)

2. Controller Package List

No.	Item	Specification
1	Main panel	☐ NEW990T ☐ NEW990M ☐ NEW1000M ☐ NEW1000T ☐ NEW1500M ☐ NEW1500T
2	Sub panel	☐ NEW1000M-A ☐ NEW1000T-A ☐ NEW1000M-B ☐ NEW1000T-B ☐ NEW1000M-C ☐ NEW1000T-C ☐ NEW1000M-E ☐ NEW1000T-E ☐ NEW1500M-A ☐ NEW1500T-A ☐ NEW1500M-B ☐ NEW1500T-B
3	Power supply	☒ RD-65B
4	CN3(I/O1)	☐ 5 meters
5	CN4(I/O3)	☐ 5 meters
6	CN5(XYZ driver signal)	☐ 5 meters
7	CN6(AB driver signal)	☐ 5 meters
8	CN8(Power input)	☐ 5 meters
9	CN9(SP-encoder input)	☐ 5 meters
10	CN10(I/O2)	☐ 5 meters
11	CN11(MPG connection)	☐ Connector ☐ MPG
12	CN13(Feedback input)	☐ 5 meters
13	CN15(Sub panel connection)	☐ 0.8 meters
14	CN16(I/O3)	☐ 5 meters
15	Relay board	☐ NEWKer-335P
16	IO Board(Bus type only)	☐ NEWKer-318E

Attention: please check package according to package list. Above items are included according to order, not all involved. if any customize requirements, please contact NEWKer staff

3. Controller connection

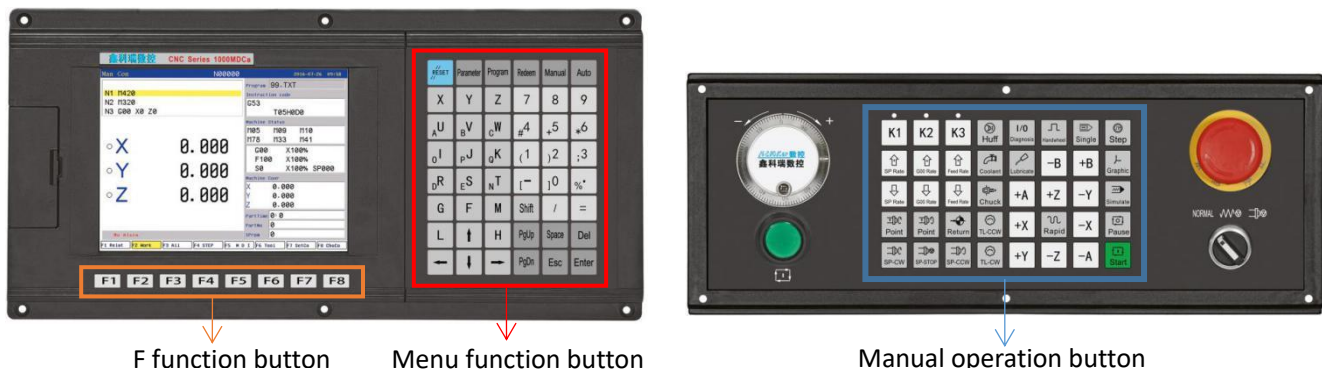


NEW18iM-3 axis connection

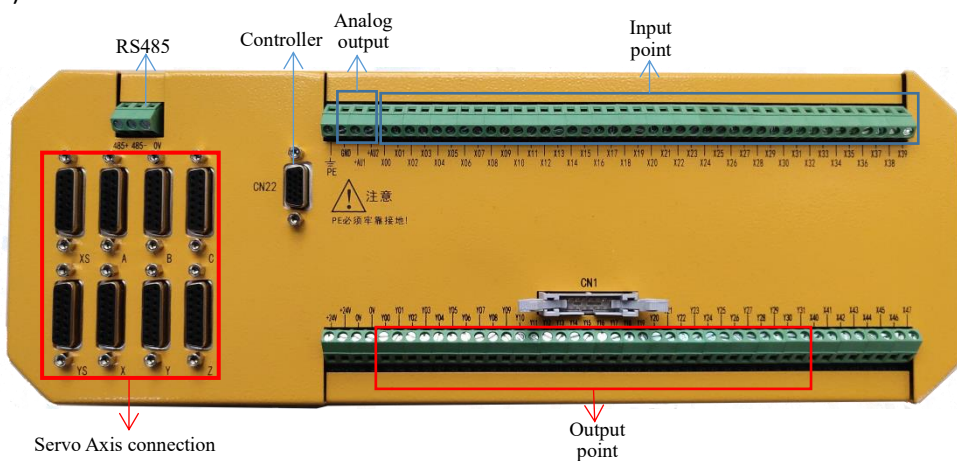
Note: Relay Board is optional, if without relay board, please connect relay according following IO chart.

4. Panel operation

A) NEW1000 series panel



B) IO Board



4. Servo axis control

1) EtherCAT control mode

Other parameter P600=4003; EtherCAT control mode

Man Con New CNC N00000 2020-12-19 16:01

Program 22
Progress

G54
T05H0D0 02T00
TrueFeed F0.000
M05 M09 M10
M78 M33 M41
G00 X100%
F500 X100%
S2000 X100% SP000
Machine Coord
X 0.000
Y 0.000
Z 0.000
PartTime 0:0
PartNo 1
SPrpm 0

400,Axis name display configur 1
401,Feed shaft pulse port conf 1
600,Control type(4000-4099:etherCAT,6000:Modbus,6001:1-6axis Modbus,6002:3axis Modbus)
:
4003
Enter Esc
611,etherCAT X Electronic gear 1
612,etherCAT Y Electronic gear 1
613,etherCAT Z Electronic gear 1

No Alarm

F1 User F2 Speed F3 Axis F4 Tool F5 Other F6 Coord F7 PASSWD F8 CANCEL

2) EtherCAT control station

Other Parameter P620=12345678; the sequence start from the first driver connecting with controller

Man Con New CNC N00000 2020-12-19 16:40

Program 22
Progress

G54
T05H0D0 02T00
TrueFeed F0.000
M05 M09 M10
M78 M33 M41
G00 X100%
F500 X100%
S2000 X100% SP000
Machine Coord
X 0.000
Y 0.000
Z 0.000
PartTime 0:0
PartNo 1
SPrpm 0

620,etherCAT Station num 12345678
621,etherCAT Servo drive 11111111
620,etherCAT Station number setting [> 10]:
12345678
Enter Esc
995,Machine tool serial number 0
996,user-defined serial number 0

No Alarm

F1 User F2 Speed F3 Axis F4 Tool F5 Other F6 Coord F7 PASSWD F8 CANCEL

3) EtherCAT driver brand

Man Con New CNC N00000 2020-12-19 16:19

Program 22
Progress

G54
T05H0D0 02T00
TrueFeed F0.000
M05 M09 M10
M78 M33 M41
G00 X100%
F500 X100%
S2000 X100% SP000
Machine Coord
X 0.000
Y 0.000
Z 0.000
PartTime 0:0
PartNo 1
SPrpm 0

620,etherCAT Station num 12345678
621,etherCAT Servo drive 11111111
621,etherCAT Servo drive type [> 1]:
11111111
Enter Esc
995,Machine tool serial number 0
996,user-defined serial number 0

No Alarm

F1 User F2 Speed F3 Axis F4 Tool F5 Other F6 Coord F7 PASSWD F8 CANCEL

Code	Driver
11111111	Maxsine/ZSMC/Invnt/Estun/STEP/Xinjie
22222222	Sanyo Denki/Panasonic A5
33333333	Enpu
44444444	Yakotec/Thinkvo/Panasonic A6/Yuhai
55555555	Bichannel Thinkvo
66666666	Tsino-Dynatron
77777777	Zhengzheng
77777773	Eura
77777774	Dorna/Kinco
77777776	DVS
88888888	Jotong/Weide

Note: if communicate failed due to scan times, please set 9th bit as 5, means delay 1 minute before communication.

5. Basic parameter setting

A) NEW18iM-3



- (a) Press enter into parameter setting, then press **F3** to go to Axis section
 (b) Electronic gear ratio setting(relative to axis movement correction):

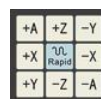
Axis parameter:

P27 X-axis electronic gear numerator = reduction ratio*10;

Man	Con	Mod	CNC	N00000	2020-12-19 16:00		
Program 22							
Progress							
G54							
T05H0D0 02T00							
TrueFeed F 0.000							
M05 M09 M10							
M78 M33 M41							
G00 X100%							
F500 X100%							
S2000 X100% SP000							
Machine Coord							
X 0.000							
Y 0.000							
Z 0.000							
PartTime 0:0							
PartNo 1							
SPrpm 0							
No Alarm							
F1 User	F2 Speed	F3 Axis	F4 Tool	F5 Other	F6 Coord	F7 PASSWD	F8 CANCEL

P28 X-axis electronic gear denominator = ball screw pitch(mm)(axis movement per turn of motor)

Man	Con	Mod	CNC	N00000	2020-12-19 16:00		
Program 22							
Progress							
G54							
T05H0D0 02T00							
TrueFeed F 0.000							
M05 M09 M10							
M78 M33 M41							
G00 X100%							
F500 X100%							
S2000 X100% SP000							
Machine Coord							
X 0.000							
Y 0.000							
Z 0.000							
PartTime 0:0							
PartNo 1							
SPrpm 0							
No Alarm							
F1 User	F2 Speed	F3 Axis	F4 Tool	F5 Other	F6 Coord	F7 PASSWD	F8 CANCEL



After setting, Press **F** on the board to set speed, then press to move X, and check if X axis real movement is real coordinate changes through dial gauge. For example, if machine coordinate changes by 10mm, dial gauge value should change by 10mm.

P29-P30 is Y axis electronic gear ratio. After setting, please also use dial gauge to check if Y real

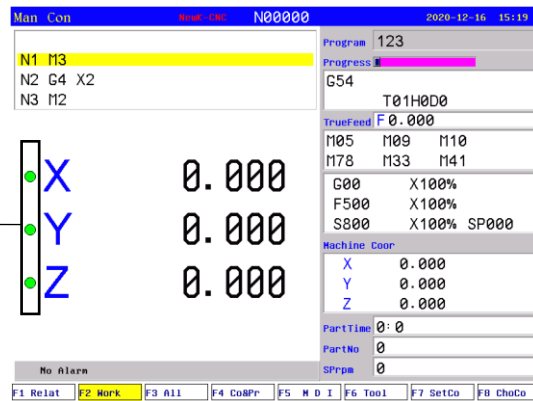
movement match with coordinate changes;

P31-P32 is Z axis electronic gear ratio. After setting, please also use dial gauge to check if Z real movement match with coordinate changes;

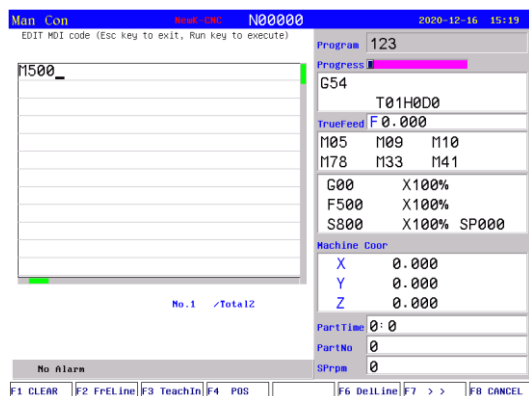
(c) Absolute driver feedback checking

If driver works well, and when controller power on, it will read position feedback from driver through Modbus.

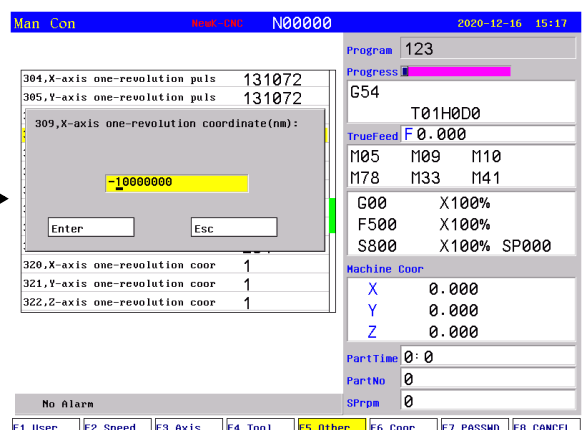
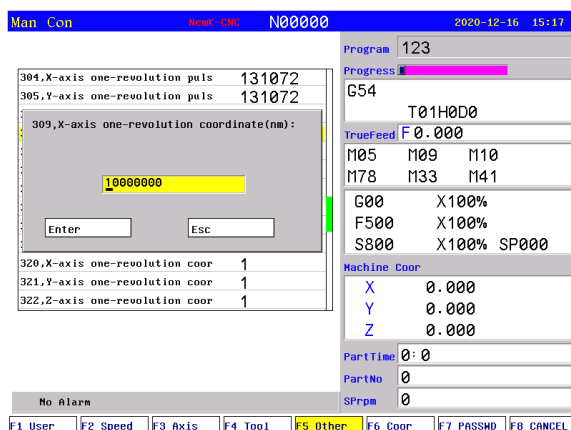
When communicate successfully, circles before coordinate will become green



1) Press **F5 MDI** to enter into MDI mode, then press **Start** to run M500 to read position ;



2) Press the keypad icon to move X by 10, then run M500 in MDI, to check if X coordinate change after run M501(change within 0.010 is acceptable). After M500, if X axis coordinate changes within 0.010, then go to step 4); if X coordinate changes over than 0.010, move X axis by 10 again, and run M500 again to check if X axis coordinate remains as before M500, if not, go to revise Other parameter P309 sign.



3) After adjust P309, run M500 in MDI again, now X coordinate will change; Move X axis by

10, and run M500, then X coordinate should remain the same as value before run M500.

4) Move Y axis by 10, run M500 in MDI to check if Y axis coordinate changes. If changes like above steps, please modify Other parameter P310; the steps are similar to Step 1-3.

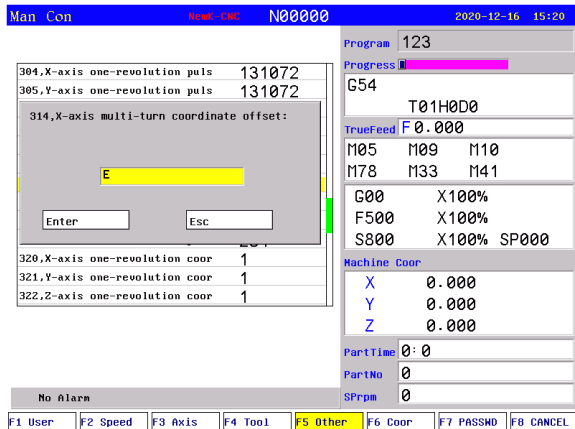
5) finish all axis feedback checking, record current machine coordinate, then restart driver with controller to check if coordinate changes before and after power on.

(d) Home process

1) Move **XYZ** to the position which will be home point of machine coordinate;

2) Set current position as home position;

Input **E** in Other parameter P314-P318, and press Enter, current XYZ machine coordinate will be 0;



3) Speed parameter

P38 X home rampit speed: X axis go home speed in positive direction;

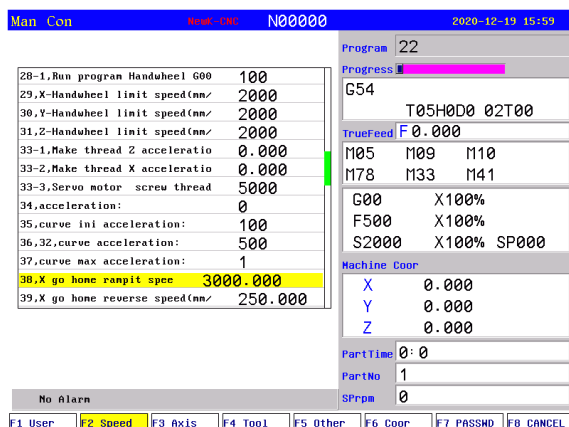
P39 X home reverse speed: X axis go home speed in negative direction;

P40 Y home rampit speed: Y axis go home speed in positive direction;

P41 Y home reverse speed: Y axis go home speed in negative direction;

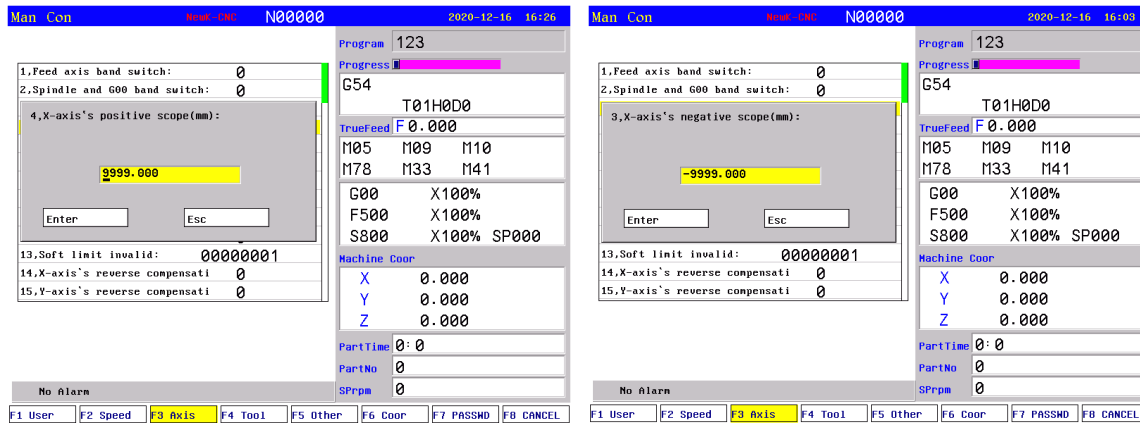
P42 Z home rampit speed: Z axis go home speed in positive direction;

P43 Z home reverse speed: Z axis go home speed in negative direction;



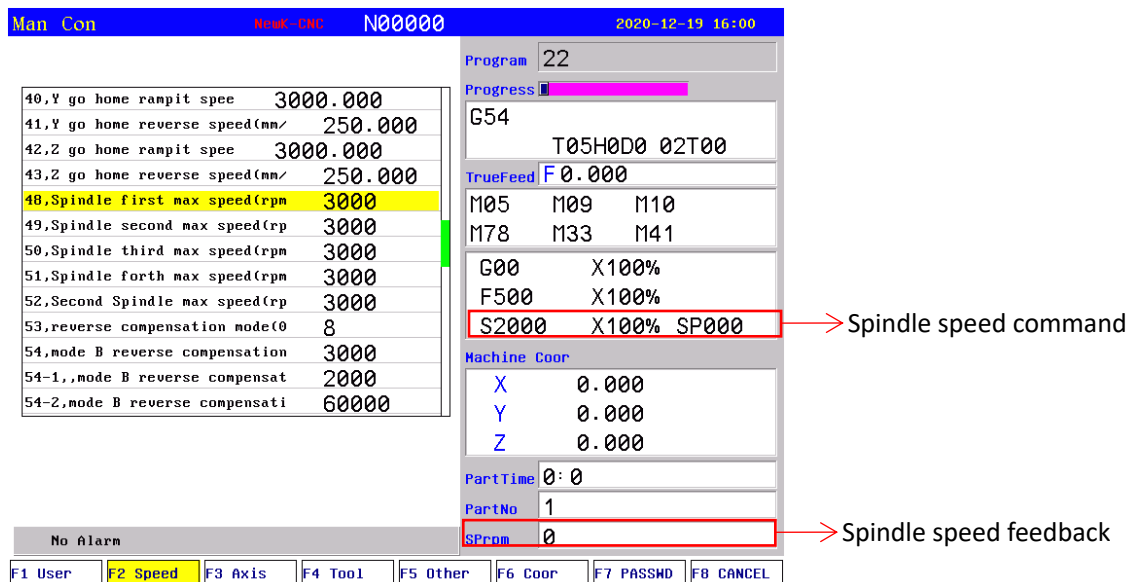
(e) Soft limit setting

Input max coordinate value in both directions into Axis parameter P3-P8;



(f) Spindle control

Item	Signal	Command	Relative Parameter	PLC address	Default I/O	Sample
1st Spindle	CW	M03	/	M53	Y14	M03 S1000
	CCW	M04	/	M54	Y13	
	Stop	M05	/	M55	Y12	
	Speed	S+speed	Speed parameter P48	/	AV1	
2nd Spindle	CW	M203	/	M213	Y28	M203 SS1000
	CCW	M204	/	M212	Y29	
	Stop	M205	/	/	/	
	Speed	SS+speed	Speed parameter P52	/	AV2	



(e) Input and Output



Press  twice to enter into IO interface:

Man Con										Head CNC										N00000										2020-12-19 17:06																																							
Input point										Output Point										Program 22																																																	
X00 T01 X01 T02 X02 T03 X03 T04 X04 T05 X05 T06 X06 T07 X07 T08 X08 A0 X09 -L X10 +L X11 Y0 X12 X0 X13 Z0 X14 / X15 / X16 YS0 X17 XS0 X18 / X19 / X20 ALM X21 TOK X22 ALM X23 ALM1 X24 ALM2 X25 M28/C0 X26 M24/B0 X27 M22 X28 M18 X29 M12 X30 M14 X31 M16 X32 / X33 / X34 / X35 / X36 / X37 / X38 / X39 / X40 / X41 / X42 / X43 / X44 / X45 / X46 / X47 /										Y00 M61 Y01 M63 Y02 M65 Y03 M67 Y04 M69 Y05 M71 Y06 M73 Y07 M59 Y08 M32 Y09 M79 Y10 M10 Y11 M08 Y12 M05 Y13 M04 Y14 M03 Y15 M75 Y16 READY Y17 INTH Y18 +T Y19 -T Y20 S04 Y21 S03 Y22 S02 Y23 S01 Y24 / Y25 / Y26 / Y27 / Y28 M203 Y29 M204 Y30 SPEN Y31 SPEN2										G54 T05H0D0 02T00 TrueFeed F0.000 M05 M09 M10 M78 M33 M41 G00 X100% F500 X100% S2000 X100% SP000 Machine Coord X 0.000 Y 0.000 Z 0.000 PartTime 0:0 PartNo 1 SPrpm 0 No Alarm																																																	
F1 Ctr1										F2 I/O										F3 LAD										F4 ALARM										F5 EdLad										F6 Reset										F8 CANCEL									

1) Input point

Signal	Command	Pin	Function	Relative parameter
X00	T01	/	Magazine tool overload alarm	/
X01	T02	/	Spindle chuck loose detection	/
X02	T03	/	Lubricant oil level detection	/
X03	T04	/	Coolant shortage alarm	/
X04	T05	/	Spindle chuck tighten detection	/
X05	T06	/	Magazine forward detection	/
X06	T07	/	Magazine backward detection	/
X07	T08	/	Tool count signal	/
X08	A0	/	A axis home signal	Axis parameter P48
X09	-L	/	Machine negative hard limit switch	Axis parameter P38
X10	+L	/	Machine positive hard limit switch	Axis parameter P37
X11	Y0	/	User-define input/Y axis home signal	Axis parameter P48
X12	X0	/	X axis home signal	Axis parameter P48
X13	Z0	/	Z axis home signal	Axis parameter P48
X14	/	/	User-define input	/
X15	/	/	User-define input	/
X16	YS0	/	Ys axis home signal	/
X17	XS0	/	Xs axis home signal	/
X18	/	/	User-define input	/
X19	/	/	User-define input	/
X20	ALM	/	XYZA axis servo alarm	Other parameter P17
X21	TOK	/	Turret lock ready	/
X22	ALM	/	BCXsYs axis servo alarm	Other parameter P17-1
X23	ALM1	/	Spindle driver alarm	Other parameter P18
X24	ALM2	/	Machine alarm	Other parameter P19
X25	M28/C0	/	User-define input/C axis home signal	/
X26	M24/B0	/	User-define input/B axis home signal	/

X27	M22	/	Spindle orientation detection	/
X28	M18	/	User-define input	/
X29	M12	/	Door switch detection	/
X30	M14	/	Compress air alarm	/
X31	M16	/	Spindle remote loosen chuck	/
X32	/	/	User-define input	/
X33	/	/	User-define input	/
X34	/	/	User-define input	/
X35	/	/	User-define input	/
X36	/	/	User-define input	/
X37	/	/	User-define input	/
X38	/	/	User-define input	/
X39	/	/	User-define input	/
X40	/	/	User-define input	/
X41	/	/	User-define input	/
X42	/	/	User-define input	/
X43	/	/	User-define input	/
X44	/	/	User-define input	/
X45	/	/	User-define input	/
X46	/	/	User-define input/M01 conditional pause	/
X47	/	/	User-define input/Spindle fan alarm	/

Attention: Common terminal of all input points is 0V, and input points are 0V effective;

Note: if controller prompt driver alarm while servo driver does not alarm, please set Other parameter P17.

2) Output point

Signal	Command	Pin	Function	Relative parameter
Y00	M61	/	Spindle orientation	/
Y01	M63	/	Magazine rotate CW	/
Y02	M65	/	Stop status light output	/
Y03	M67	/	Alarm status light output	/
Y04	M69	/	Run status light output	/
Y05	M71	/	Magazine rotate CCW	/
Y06	M73	/	Magazine forward	/
Y07	M59	/	Huff	/
Y08	M32	/	Lubricant	Other parameter P4-P6
Y09	M79	/	User-define Output	/
Y10	M10	/	Spindle chuck	Other parameter P13, P20,P22,P24
Y11	M08	/	Coolant	Other parameter P14
Y12	M05	/	Spindle stop	Axis parameter P13-P14
Y13	M04	/	Spindle CCW	/
Y14	M03	/	Spindle CW	/
Y15	M75	/	Spindle control mode switch	/
Y16	READY	/	Servo enable(default close)	/
Y17	INTH	/	Servo alarm reset	/

Y18	+T	/	User-define Output	/
Y19	-T	/	User-define Output	/
Y20	S04(M44)	/	Spindle 4th gear	Speed parameter P51
Y21	S03(M43)	/	Spindle 3rd gear	Speed parameter P50
Y22	S02(M42)	/	Spindle 2nd gear	Speed parameter P49
Y23	S01(M41)	/	Spindle 1st gear	Speed parameter P48
Y24	/	/	User-define Output	/
Y25	/	/	User-define Output	/
Y26	/	/	User-define Output	/
Y27	/	/	User-define Output	/
Y28	M203	/	2nd spindle CW	/
Y29	M204	/	2nd spindle CCW	/
Y30	SPEN	/	1st spindle Enable	/
Y31	SPEN2	/	2nd spindle Enable	/

Attention: Common terminal of all output points is 24V, and output points are 0V effective;

Sample: when run M61, Y00 will be on; when run M60, Y00 will be off.

NEWKER数控 成都鑫科瑞数控技术有限公司
鑫科瑞数控 CHENGDU NEWKER CNC-TECHNOLOGY CO., LTD

Address: Economic Headquarter E2, No.888 Chenglong Avenue, Longquanyi District, Chengdu City, Sichuan Province, People's Republic of China.

Website: WWW.NEWKER-CNC.COM

Email: NEWKER.CNC@aliyun.com

Phone: 028-84601148

Fax: 028-84609226

Skype: Newker-CNC James

Whatsapp: +8618990029555